AMReX/incflo

Open Source CFD Solver

What is incflo?

- Solves for the velocity and pressure of an incompressible fluid
- 3D Objects can be inserted to simulate how this affects flow
- Numerically solves the Navier-Stokes equations
- Developed primarily at Lawrence Berkeley National Laboratory
- CFD is used primarily by aerospace and mechanical engineers

```
 ============   NEW TIME STEP   ============

Step 2: from old_time 9.803319611e-05 to new time 0.0002058697118 with dt = 0.0001078365157.

MAC Projection:
MLMG: Initial rhs               = 19.01866657
MLMG: Initial residual (resid0) = 19.01866657
MLMG: Final Iter. 8 resid, resid/bnorm = 1.264739082e-11, 6.649988197e-13
MLMG: Timers: Solve = 0.091577498 Iter = 0.09068665 Bottom = 0.066768348
REDISTRIBUTION TYPE StateRedist
Diffusing scalars one at a time ...
Diffusing velocity components all together...
Nodal Projection:
 >> Before projection:
  * On lev 0 max(abs(rhs)) = 4.713718554

MLMG: Initial rhs               = 4.713718554
MLMG: Initial residual (resid0) = 4.713718554
MLMG: Final Iter. 7 resid, resid/bnorm = 1.781086389e-11, 3.778516619e-12
MLMG: Timers: Solve = 0.199979274 Iter = 0.174914416 Bottom = 0.092704964
 >> After projection:
  * On lev 0 max(abs(rhs)) = 1.907279798

Time per step 0.465298752
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AMReX (26.11-9-g65c32d1437f2-dirty) initialized
incflo git hash: 24.11-65-g18045d0
Scalar diffusion coefficients
Tracer diffusion coeff: 0:0.001
Newtonian fluid with mu = 0.001
incflo.geometry_filename:

 Building cylinder geometry.

 Internal Flow: 0
 Radius:    0.05000001
 Direction: 2
 Rotation angle(rad): 0
 Rotation axe: 0
 Center:    0.151, 0.2, 0
Done making the EB geometry index space.

xlo set to mass inflow.
xhi set to pressure outflow.
ylo set to no-slip wall.
yhi set to no-slip wall.
Making new level 0 from scratch
Nodal Projection:
 >> Before projection:
  * On lev 0 max(abs(rhs)) = 114.1960446

MLMG: Initial rhs                = 114.1960446
MLMG: Initial residual (resid0) = 114.1960446
MLMG: Final Iter. 8 resid, resid/bnorm = 2.070663641e-10, 1.813253382e-12
MLMG: Timers: Solve = 0.234327572 Iter = 0.204002721 Bottom = 0.109350981
 >> After projection:
  * On lev 0 max(abs(rhs)) = 5.26317442

 In initial_iterations: iter = 0
MAC Projection:
MLMG: Initial rhs                = 17.4100553
MLMG: Initial residual (resid0) = 17.4100553
MLMG: Final Iter. 8 resid, resid/bnorm = 3.051092746e-11, 1.75248883e-12
MLMG: Timers: Solve = 0.104555703 Iter = 0.103629827 Bottom = 0.079533756
REDISTRIBUTION TYPE StateRedist
Diffusing scalars one at a time ...
Diffusing velocity components all together...
Nodal Projection:
 >> Before projection:
  * On lev 0 max(abs(rhs)) = 3.097821272

MLMG: Initial rhs                = 3.097821272
MLMG: Initial residual (resid0) = 3.097821272
MLMG: Final Iter. 8 resid, resid/bnorm = 1.079525358e-12, 3.484789028e-13
MLMG: Timers: Solve = 0.23236309 Iter = 0.204799455 Bottom = 0.110254203
 >> After projection:
  * On lev 0 max(abs(rhs)) = 0.694557497


 In initial_iterations: iter = 1
MAC Projection:
MLMG: Initial rhs                = 17.0534988
MLMG: Initial residual (resid0) = 17.0534988
MLMG: Final Iter. 8 resid, resid/bnorm = 2.247431374e-11, 1.31787113e-12
MLMG: Timers: Solve = 0.102743808 Iter = 0.101900225 Bottom = 0.077864108
REDISTRIBUTION TYPE StateRedist
Diffusing scalars one at a time ...
Diffusing velocity components all together...
Nodal Projection:
 >> Before projection:
  * On lev 0 max(abs(rhs)) = 0.7504786532

MLMG: Initial rhs                = 0.7504786532
MLMG: Initial residual (resid0) = 0.7504786532
MLMG: Final Iter. 7 resid, resid/bnorm = 6.101785743e-13, 8.130525388e-13
MLMG: Timers: Solve = 0.199266335 Iter = 0.171253188 Bottom = 0.08808964
 >> After projection:
  * On lev 0 max(abs(rhs)) = 0.4254754583
```

What makes incflo different?

- Focuses on variable density fluids
  - Focus on fluid mixing
- Uses cut-cells to model 3D objects
  - Computationally less expensive than importing a mesh
- Uses Adaptive Mesh Refinement (AMR)
  - Adds more cells where there’s complex geometry

How does the algorithm work?

- Calculations are performed iteratively over multiple time steps
- For each time step:
  - **Step 1:** Calculator velocity using Navier-Stokes without pressure term
  - **Step 2:** Use a linear system to calculate the pressure in each cell
  - **Step 3:** Correct velocity using pressure term

Step 1: Provisional Velocity

Full Momentum Equation

$$\frac{\partial \mathbf{U}}{\partial t} = -(\mathbf{U} \cdot \nabla)\mathbf{U} - \frac{1}{\rho}\nabla p + \frac{1}{\rho}\nabla \cdot \boldsymbol{\tau} + \frac{\mathbf{f}}{\rho}$$

Equation of each time step

$$\frac{\mathbf{U}^* - \mathbf{U}^n}{\Delta t} = -(\mathbf{U}^n \cdot \nabla)\mathbf{U}^n + \frac{1}{\rho^n}\nabla \cdot \boldsymbol{\tau}^n + \mathbf{g} + \frac{\mathbf{F}^n}{\rho^n}$$

- Gradients computed by performing calculations with velocity of neighbors
- Program runs calculations in parallel to optimize performance

Step 2: Calculate Pressure

Using the below equation,we can solve for the pressure (ϕ) at each cell.

$$\nabla \cdot \left(\frac{1}{\rho(\mathbf{x})} \nabla \phi(\mathbf{x}) \right) = \nabla \cdot \mathbf{U}^*(\mathbf{x})$$

- We can turn the above equation in the form Ax = b to solve as a linear equation.
- Linear system is adjusted to account for the size of cell and amount of fluid in cell
- Uses Multi-Level Multi-Grid (MLMG) solver to dampen errors across course and fine grid hierarchies to achieve O(n)
- Uses local cell data instead of matrices to perform calculations

Step 3: Correct Velocity

Use the calculated pressure to correct the provisional velocity

$$\mathbf{U}^{n+1} = \mathbf{U}^{*} - \frac{\Delta t}{\rho}\nabla\phi$$

Accuracy and Conditioning

- For some test simulations, numerical outputs are compared to known analytical solutions at each time step
- This solver is mean to be more accurate since it accounts for variable density
- Extremely small cut cells or complex geometries can lead to ill-conditioning
- The software will redistribute how smaller cells are calculated to reduce ill-conditioning
- MLMG helps to avoid ill conditioning of extremely large matrices

Question and Experiment

- Question: What algorithms and computational expenses are involved in creating the cut cells?
- Experiment: How does the speed of a fluid vary over time for a particular point depending on the inserted geometry?